The Dataverse Project

You Are Not Your User

Dispelling Usability Myths with Harvard's Dataverse

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Institute for Quantitative Social Science
Harvard University

dataverseorg

Session Description

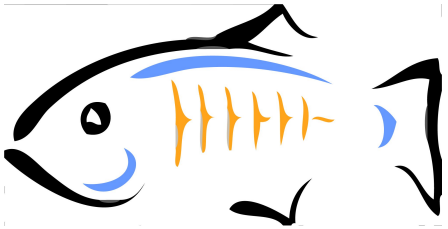
All of us have experience in developing Java applications, but how many of us know how best to make them usable? UX is a team effort, not just one person's role. This presentation goes over some common UX myths and how a developer and a UXer can work together to improve the UX of an application. Reviewing examples from the development effort of Harvard's Dataverse Project, it shows how this project has dispelled some of these myths and solved them with Java techniques. It also shares some simple ways to incorporate UX practices into your product development cycle.

Dataverse Team

- Began in 2006
- Number of developers has varied over years
- Currently:
  - 4 dedicated developers
  - 1 lead
  - shared QA
  - shared UX specialist and UI designer
- Initially, involved internal users for UI design /reviews
- In 2013, hired students from Simmons College to do a Usability Review
  - afterwards, brought UX specialist (Elizabeth) on full time

Dataverse Technology

Glassfish Server 4.1



Java SE7

- plan to upgrade to Java SE8

Java EE7

- Presentation: JSF (PrimeFaces), RESTful API
- Business: EJB, Transactions, Asynchronous, Timers
- Storage: JPA (Entities), Bean Validation

Software framework for publishing, citing and preserving research data
(open source on github for others to install)

Provides incentives for researchers to share:

- Recognition & credit via data citations
- Control over data & branding
- Fulfill Data Management Plan requirements



What is a Dataverse or Dataset?

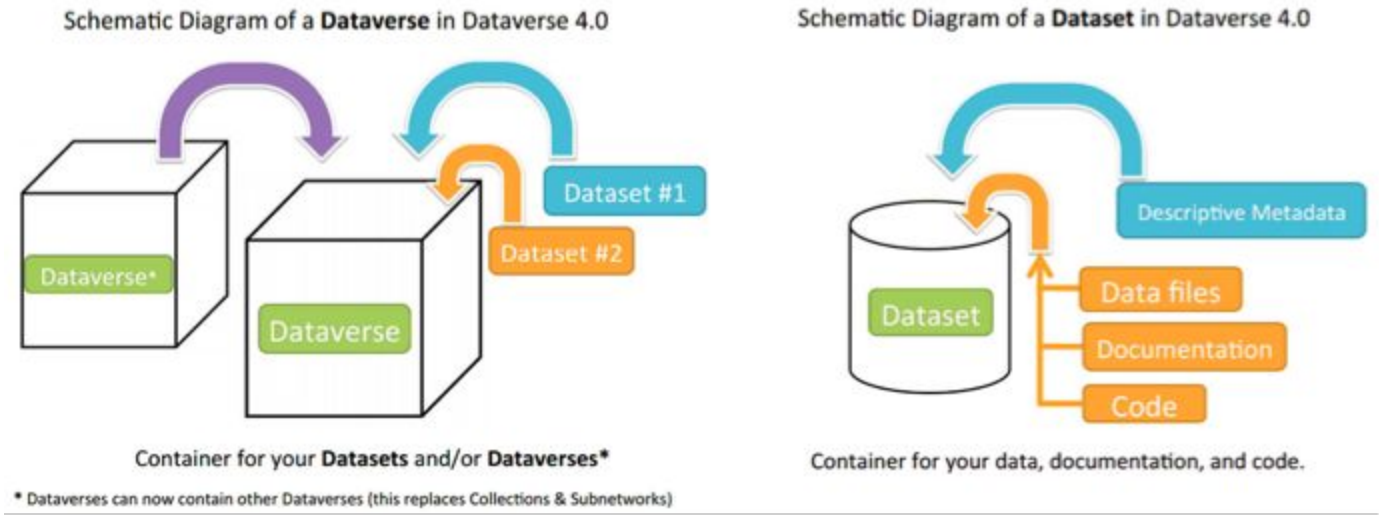


Image created by: Eleni Castro

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What are UX myths?

“UX Myths collects the most frequent user experience misconceptions and explains why they don't hold true.” - uxmyths.com

Created by Zoltan Gocza and Zoltan Kollin

“Our goal is to provide evidence in user experience design that can help stakeholders move away from design decisions that are based merely on beliefs and personal opinions. But you should still do your own research, check how your design performs.”

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UX design is a step in a project

&

UX design is about usability

UX Design is a step in a project

Many think that user experience design is confined to sketching the interfaces. However, UX design is a much broader process that - ideally - ***starts at the strategy level and affects the whole lifecycle of a project or a business.***

UX Design is about usability

Usability allows people to ***easily accomplish their goals***. ***UX design*** covers more than that, it's about ***giving people a delightful and meaningful experience***.

A good design is pleasurable, thoughtfully crafted, makes you happy, and gets you immersed.

How UX solves problems

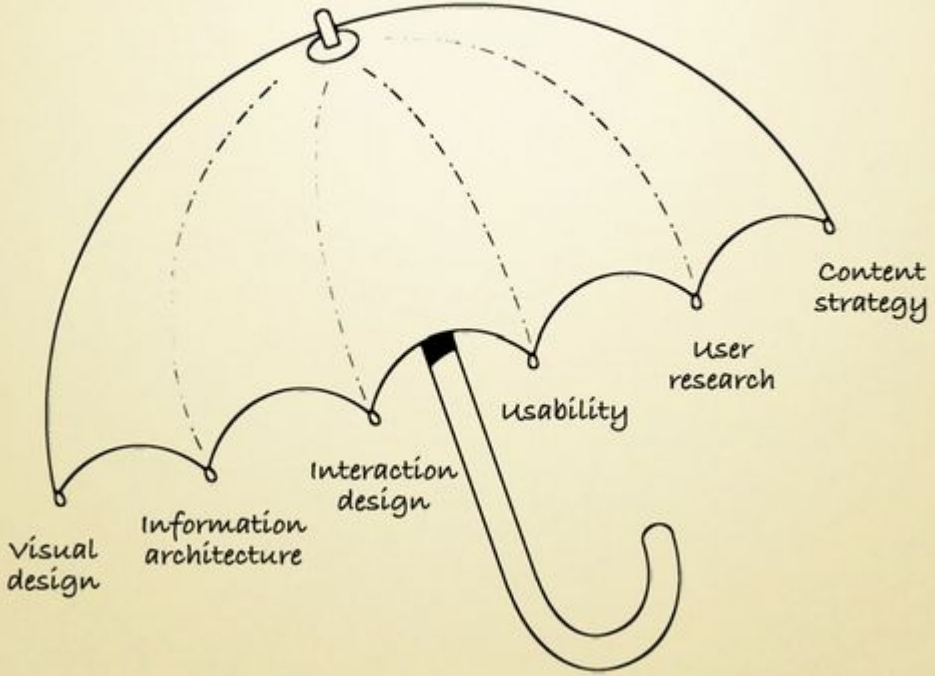


Image by Dan Willis (@uxcrank)



https://twitter.com/lukew



Luke Wroblewski
@lukew

priorities:
1. useful
2. usable
3. desirable

somethings never change.

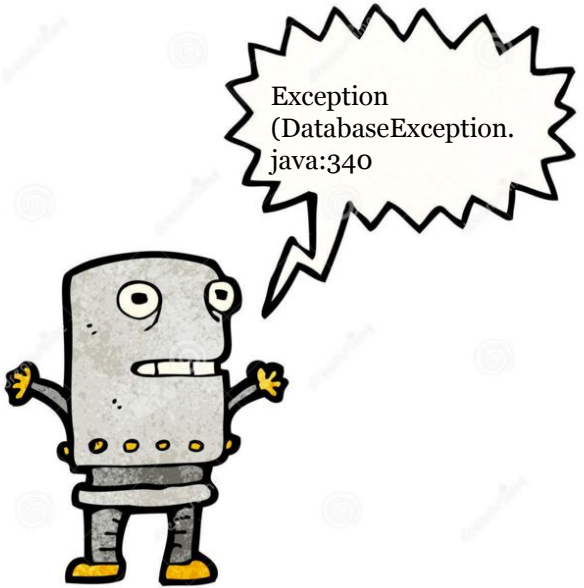
Functional Requirements Document

UI Workflows:

- Log In->Account Page->Data Related to Me Tab
- Log In->Account Page->Data Related to Me Tab->click on facet to narrow down to only dataverses
- Log In->Account Page->Data Related to Me Tab->click on facet to only see datasets
- Log In->Account Page->Data Related to Me Tab->click on facet to only see files
- Log In->Account Page->Data Related to Me Tab->Add Data->New Dataverse->Pop up saying, "This dataverse will be created under the Harvard Dataverse."
- Log In->Account Page->Data Related to Me Tab->Add Data->New Dataset->Pop up saying, "This dataset will be created under the Harvard Dataverse."

The importance of content

- Be open to changing terminology, labels, and help text
- Be aware of robot terminology in UI
- Have a connected experience:
  - Links to guides in the application
  - Don’t make users hunt for what they are looking for



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Usability testing is expensive

In fact, usability tests can be ***both fast and relatively cheap.*** You ***don't need expensive prototypes; low-tech paper prototype tests*** can also bring valuable results. You don't need a lot of participants either, even 5 users can be enough to test for specific tasks, and the recruiting can also be done guerilla-style. For many projects, ***you can even use remote and unmoderated tests.***

Remote Moderated Usability Testing

- Connect with users around the world
- Reduces cost of incentives
- Easier to manage
- Shows me how our products work on other computers and browsers

Remote Moderated Usability Testing Tools

For connecting:

- Google Hangouts
- Skype
- join.me
- Go to Meeting

For recording:

- Morae
- Quicktime
- Silverback

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Usability testing = focus groups

When it comes to collecting feedback from users, ***usability tests and focus groups are often confused although their goals are completely different.***

Focus groups assess what users say: a number of people gather in order to discuss their feelings, attitudes and thoughts on a given topic to reveal their motivations and preferences.

Usability testing, on the other hand, is ***about observing how people actually use a product***, by assigning key tasks to users and analyzing their performance and experience.

But what if your boss or the stakeholder doesn’t want usability testing done?

Contextual Inquiries

- No scenario or tasks planned

- Takes place in the user's office, home, wherever they are comfortable and where they use the product most

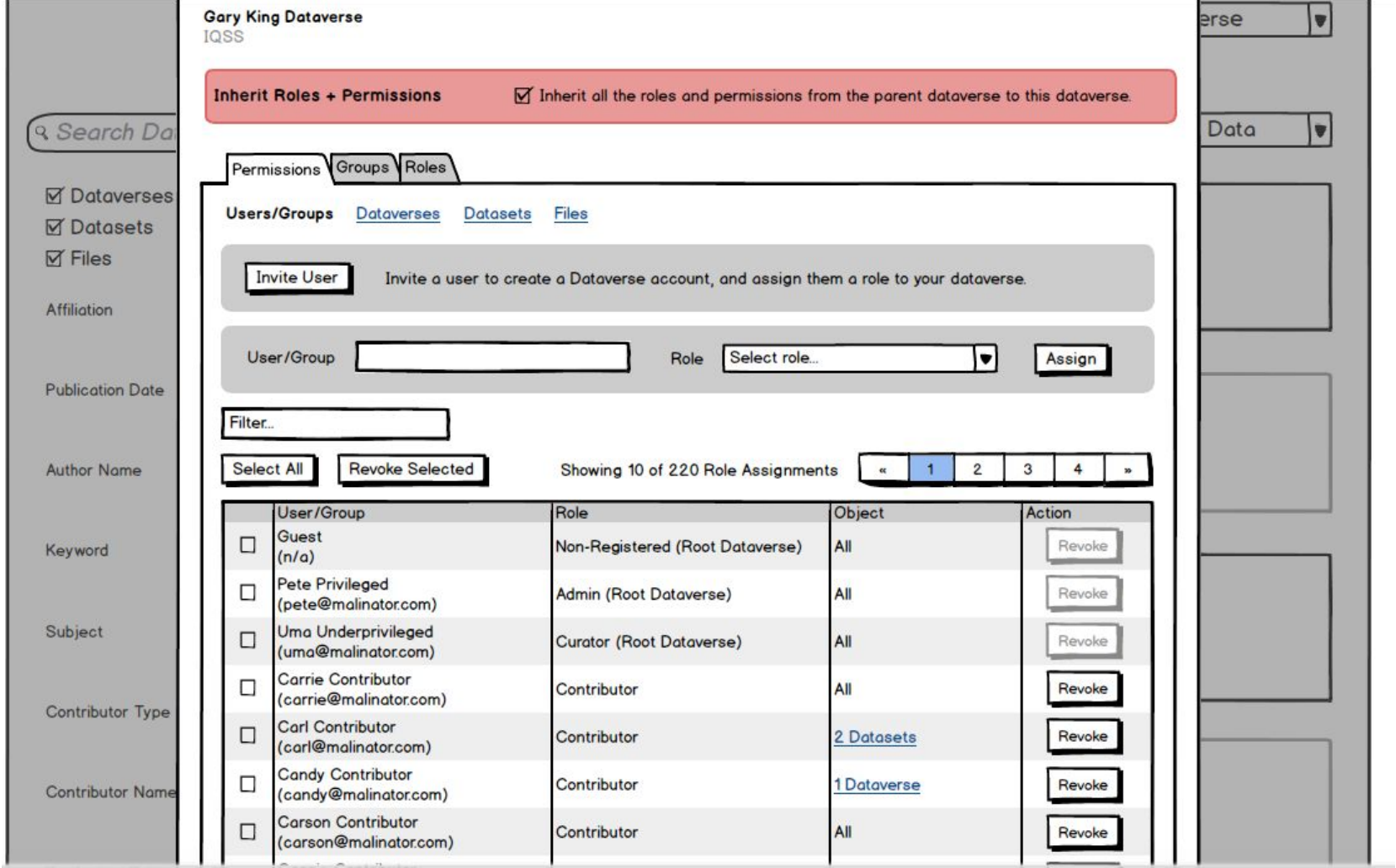
- Sit there and watch the user interact with the product as they complete tasks they normally do

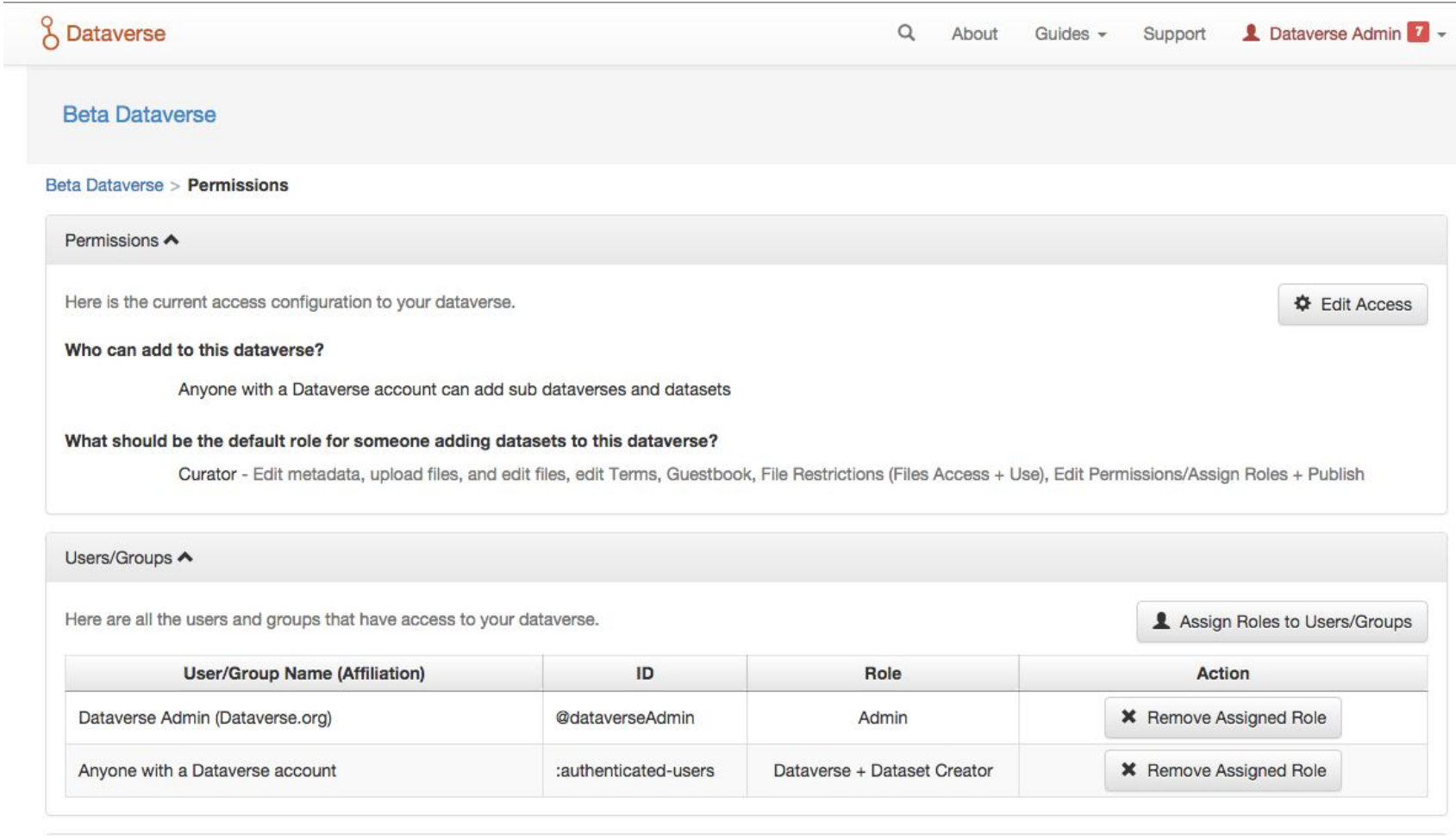


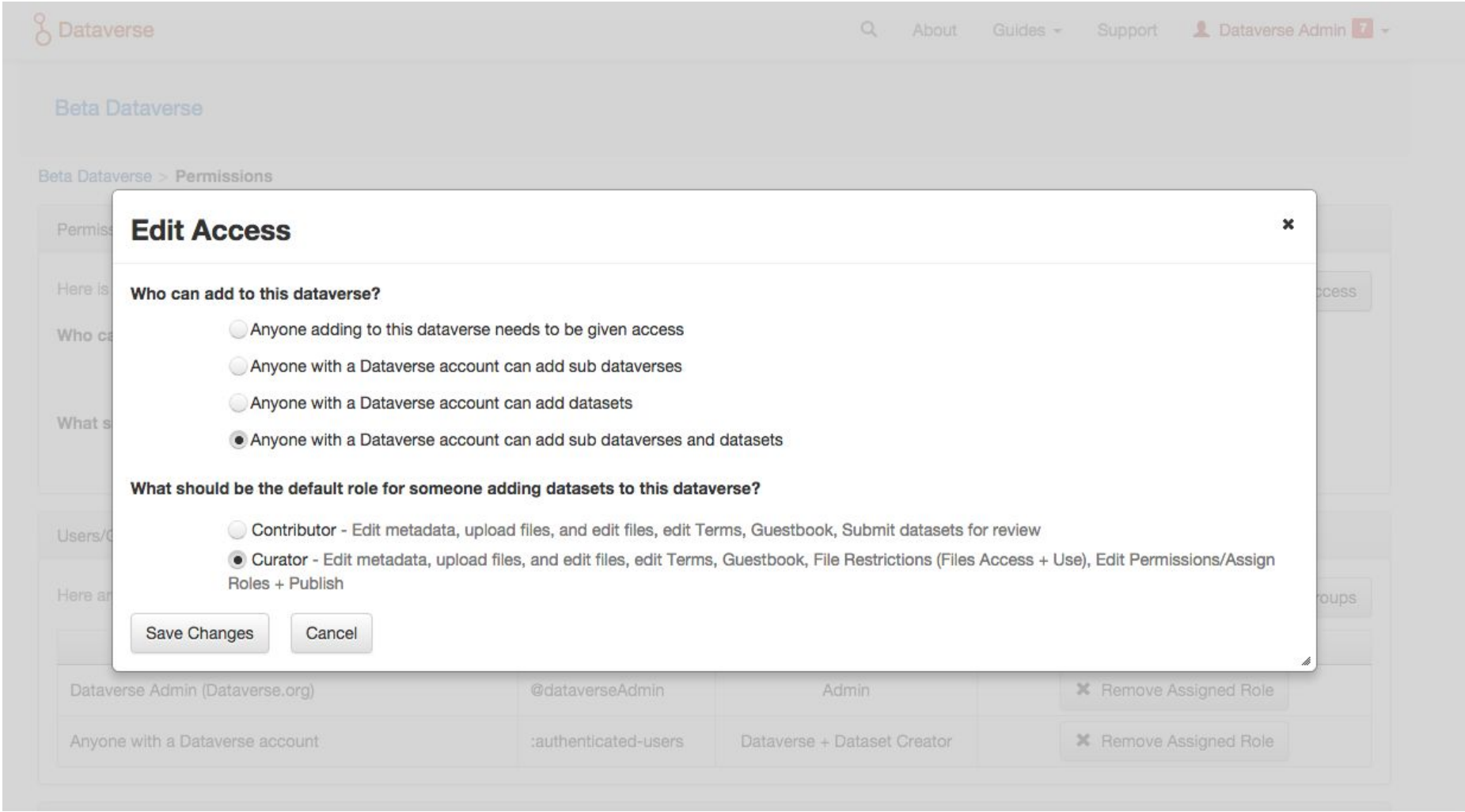
If you are an expert, you don’t need to test your design

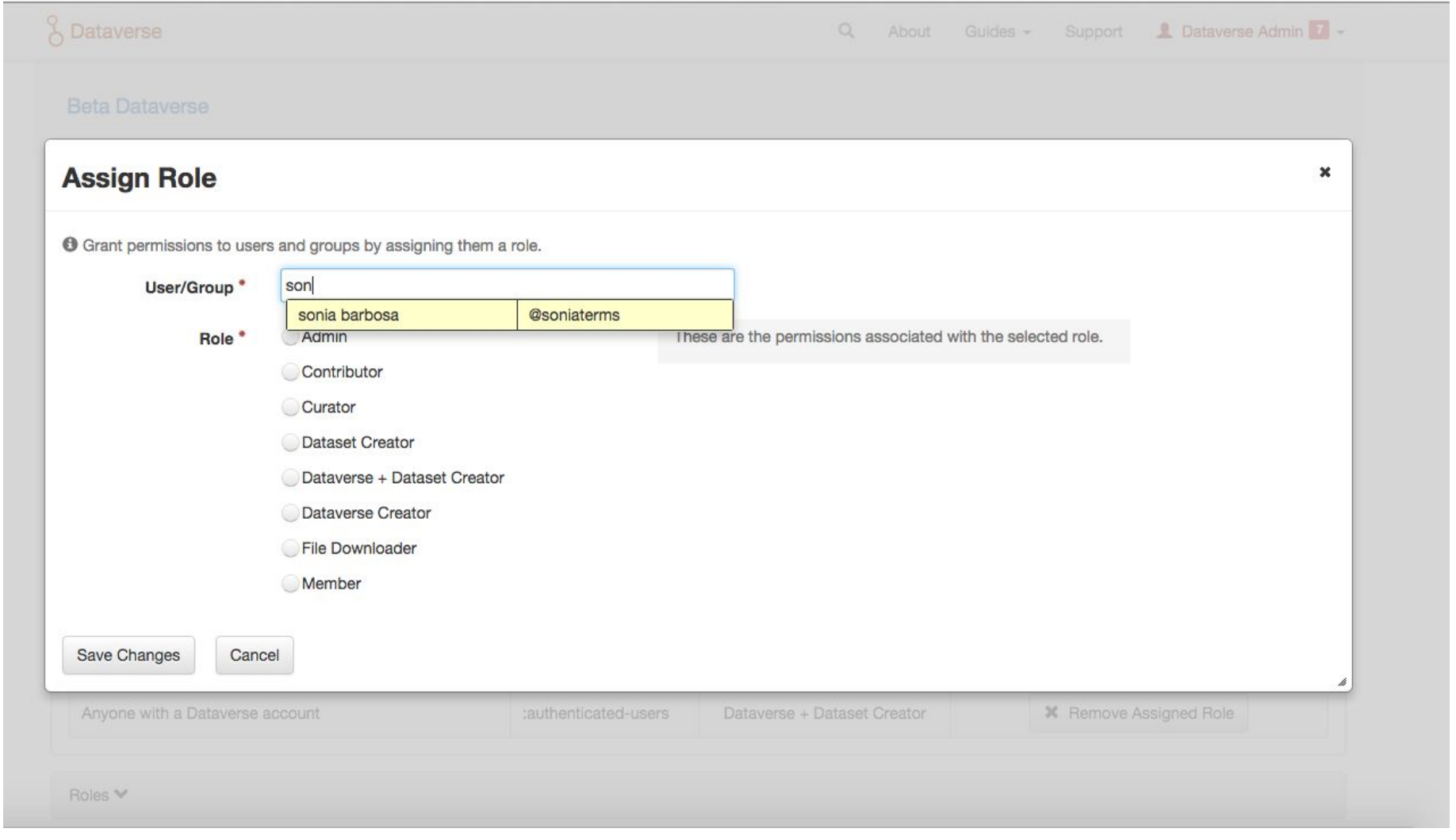
When it comes to evaluating the usability of an interface, user testing is often considered unnecessary if an expert has already reviewed it. Since people rarely behave the way you expect, ***an expert can find major usability problems, but usability tests always reveal surprising issues.***

Usability testing and expert reviews are both useful and tend to have different findings, therefore it's usually recommended to combine the two in order to get the most comprehensive analysis of the interface.


Gary King Dataverse
IQSS
Inherit Roles + Permissions
Inherit all the roles and permissions from the parent dataverse to this dataverse.
Permissions
Groups
Roles
Users/Groups
Dataverses
Datasets
Files
Invite User
Invite a user to create a Dataverse account, and assign them a role to your dataverse.
User/Group
Role
Select role...
Assign
Filter...
Select All
Revoke Selected
Showing 10 of 220 Role Assignments
« 1 2 3 4 »
User/Group
Role
Object
Action
Guest (n/a)
Non-Registered (Root Dataverse)
All
Revoke
Pete Privileged (pete@malinator.com)
Admin (Root Dataverse)
All
Revoke
Uma Underprivileged (uma@malinator.com)
Curator (Root Dataverse)
All
Revoke
Carrie Contributor (carrie@malinator.com)
Contributor
All
Revoke
Carl Contributor (carl@malinator.com)
Contributor
2 Datasets
Revoke
Candy Contributor (candy@malinator.com)
Contributor
1 Dataverse
Revoke
Carson Contributor (carson@malinator.com)
Contributor
All
Revoke
Search Da
Dataverses
Datasets
Files
Affiliation
Publication Date
Author Name
Keyword
Subject
Contributor Type
Contributor Name
Data


Dataverse
About
Guides
Support
Dataverse Admin 7
Beta Dataverse
Beta Dataverse > Permissions
Permissions
Here is the current access configuration to your dataverse.
Edit Access
Who can add to this dataverse?
Anyone with a Dataverse account can add sub dataverses and datasets
What should be the default role for someone adding datasets to this dataverse?
Curator - Edit metadata, upload files, and edit files, edit Terms, Guestbook, File Restrictions (Files Access + Use), Edit Permissions/Assign Roles + Publish
Users/Groups
Here are all the users and groups that have access to your dataverse.
Assign Roles to Users/Groups
User/Group Name (Affiliation)
ID
Role
Action
Dataverse Admin (Dataverse.org)
@dataverseAdmin
Admin
Remove Assigned Role
Anyone with a Dataverse account
:authenticated-users
Dataverse + Dataset Creator
Remove Assigned Role


Dataverse
About
Guides
Support
Dataverse Admin
Beta Dataverse
Beta Dataverse > Permissions
Edit Access
Who can add to this dataverse?
Anyone adding to this dataverse needs to be given access
Anyone with a Dataverse account can add sub dataverses
Anyone with a Dataverse account can add datasets
Anyone with a Dataverse account can add sub dataverses and datasets
What should be the default role for someone adding datasets to this dataverse?
Contributor - Edit metadata, upload files, and edit files, edit Terms, Guestbook, Submit datasets for review
Curator - Edit metadata, upload files, and edit files, edit Terms, Guestbook, File Restrictions (Files Access + Use), Edit Permissions/Assign Roles + Publish
Save Changes
Cancel
Dataverse Admin (Dataverse.org)
@dataverseAdmin
Admin
Remove Assigned Role
Anyone with a Dataverse account
:authenticated-users
Dataverse + Dataset Creator
Remove Assigned Role


Dataverse
About
Guides
Support
Dataverse Admin
Beta Dataverse
Assign Role
Grant permissions to users and groups by assigning them a role.
User/Group *
son
sonia barbosa
@soniaterms
Role *
These are the permissions associated with the selected role.
Admin
Contributor
Curator
Dataset Creator
Dataverse + Dataset Creator
Dataverse Creator
File Downloader
Member
Save Changes
Cancel
Anyone with a Dataverse account
:authenticated-users
Dataverse + Dataset Creator
Remove Assigned Role
Roles

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Design has to be original

Many designers would rather attempt reinventing the wheel than to adapt conventional user interface design patterns. It should be considered, however, that such ***design conventions are well-working because they've already been introduced and tested for usability.*** Since the users know them well, you don't need any explanation or instruction manual. As users appreciate usability over novelties, ***standard patterns will eventually benefit your audience.***

It might occur that a new approach is needed, but you must be 100% positive that your solution is better than the existing pattern.

Search this dataverse... | Find | Advanced Search | + Add Data

- [x] Dataverses (1,303)
- [x] Datasets (59,396)
- [ ] Files (285,406)

Dataverse Category

Researcher (378)
Research Project (170)
Organization or Institution (126)
Journal (60)
Teaching Course (8)

Publication Date

2015 (15,128)
2011 (10,074)
2007 (9,586)
2012 (8,642)
2009 (6,251)

More...

Subject

Social Sciences (5,186)
Earth and Environmental Sciences (249)
Medicine, Health and Life Sciences (226)
Other (179)
Computer and Information Science (107)

More...

Author Name

Digital Archive of Massachusetts Anti-Slavery

1 to 10 of 60,699 Results | Sort

Replication Data to Effect of Subliminal Lexical Priming on the Subjective Perception of Images: a Machine Learning Approach
Oct 29, 2015

Menoth Mohan, Dhanya; Kumar, Parmod; Mahmood, Faisal; Foong Wong, Kian; Agrawal, Abhishek; Elgendi, Mohamed; Shukla, Rohit; Ang, Natania; Ching, April; Dauwels, Justin; Chan, Alice H.D., 2015, "Replication Data to Effect of Subliminal Lexical Priming on the Subjective Perception of Images: a Machine Learning Approach", http://dx.doi.org/10.7910/DVN/VIWTIQ, Harvard Dataverse, V1

The purpose of the study is to examine the effect of subliminal priming in terms of the perception of images influenced by words with positive, negative, and neutral emotional content, through electroencephalograms (EEGs). Participants were instructed to rate how much they like t...

Reproduction files for "The Generalizability of Survey Experiments"
Oct 29, 2015 - Thomas J. Leeper Dataverse

Mullinix, Kevin J.; Leeper, Thomas J.; Druckman, James N.; Freese, Jeremy, 2015, "Reproduction files for "The Generalizability of Survey Experiments"", http://dx.doi.org/10.7910/DVN/MUJHGR, Harvard Dataverse, V1

This archive contains datasets, R analysis files, and supplemental materials for: Mullinix, Kevin J., Thomas J. Leeper, James N. Druckman, and Jeremy Freese. "The Generalizability of Survey Experiments." Journal of Experimental Political Science, Forthcoming. See README for detai...

Mau Forest Household baseline
Oct 29, 2015 - Nile-Congo Dataverse

Baraka, Paul; Okia, Clement; Chiputwa, Brian; Gassner, Anja; Makui, Permutia, 2015, "Mau Forest Household baseline", http://dx.doi.org/10.7910/DVN/3PEMLB, Harvard Dataverse, V1

Household baseline data collected in the Mau sentinel site.

Nile-Congo Dataverse (World Agroforestry Centre)
Oct 29, 2015 Sentinel Landscapes Dataverse

This dataverse holds data collected at the Household and Village level of the Nile-Congo sentinel site.


Do not want.

But what if you don’t have standards to follow?

United States Digital Services + 18F

- Two groups in the U.S. government working towards improving websites and applications produced by the U.S. government

- Jointly created the U.S. Web Design Standards
  - Includes code available on Github to download for UI components

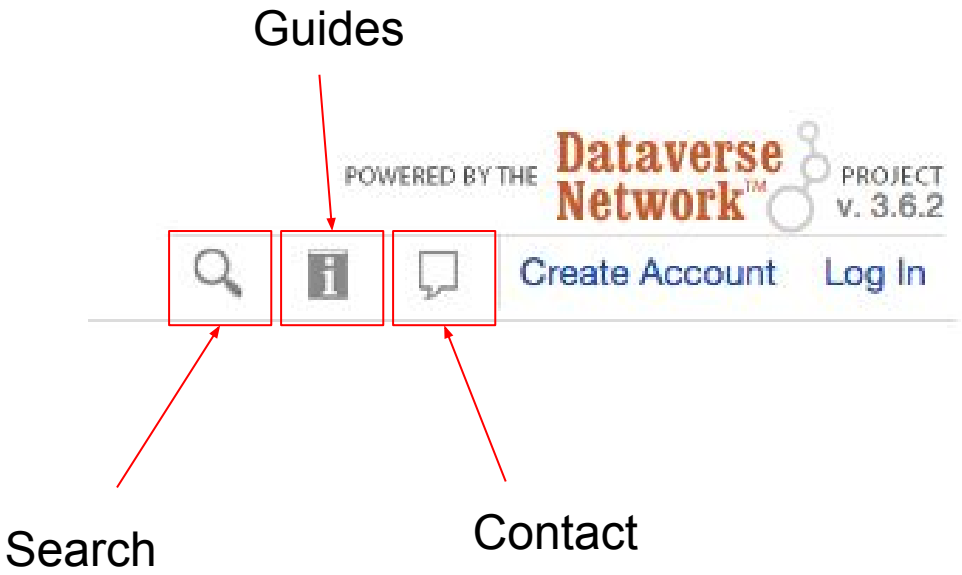
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Icons enhance usability

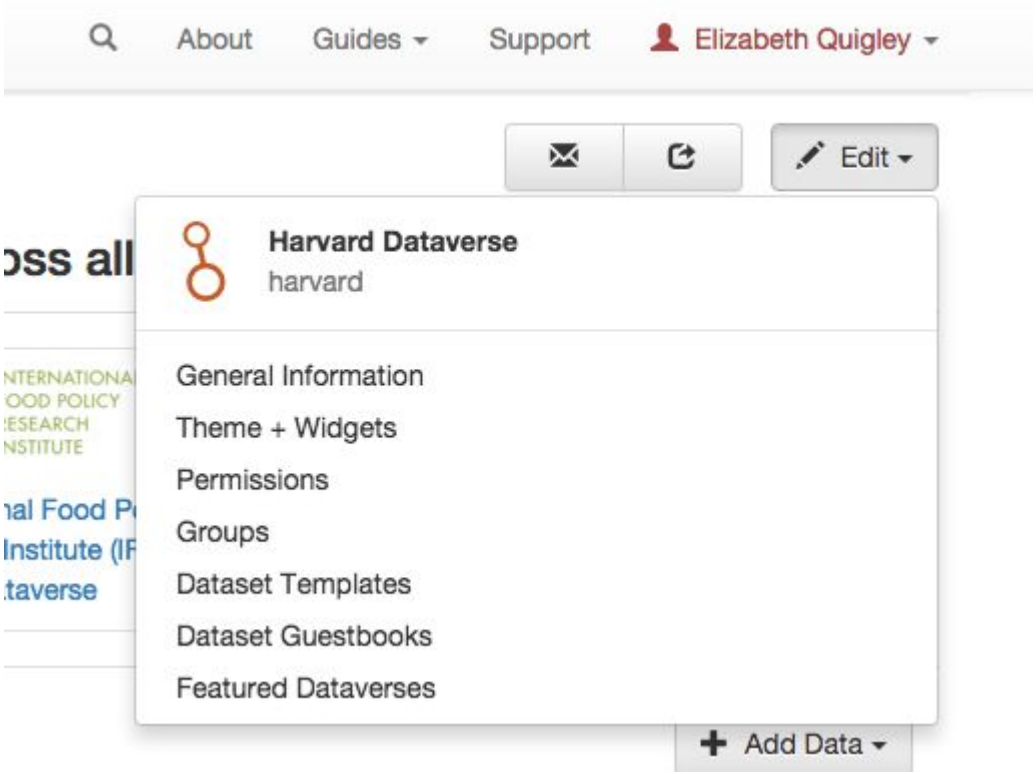
Many researchers have shown that ***icons are hard to memorize and are often highly inefficient.*** The Microsoft Outlook toolbar is a good example: the former icon-only toolbar had poor usability and changing the icons and their positioning didn't help much. What did ***help was the introduction of text labels next to the icons***. It immediately fixed the usability issues and people started to use the toolbar. In another study, the team of UIE observed that ***people remember a button's position instead of the graphic interpretation of the function.***

In most projects, icons are very difficult to get right and need a lot of testing. For abstract things, icons rarely work well.

Dataverse v3.6.2



Dataverse v4



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People always use your product the way you imagined they would

Even if a product was ***designed to fulfill specific and known user needs, customers don't always use it the way and for the purpose the product was originally intended.***

In many cases, ***users don't care or don't understand how a product works,*** and once they find a way to use it, they'll stick to it. Many people, for example, type URLs into the Google search bar instead of the browser's address bar.

You should, therefore, ***never take your design for granted and always collect feedback on how your product is actually used to reveal the real user needs and to get ideas of innovation.***

Description For testing file scrolling feature

Subject Computer and Information Science

Files | Metadata | Terms | Versions

Search this dataset... Find

1,000 Files Download

MD5: a13387af5ff7edadf46c550c60d2437f;

file829.txt
Plain Text - 28 bytes - Oct 25, 2015 - 0 Downloads
MD5: fdf949102f4d72c960b7b863cacc7e8f; Download

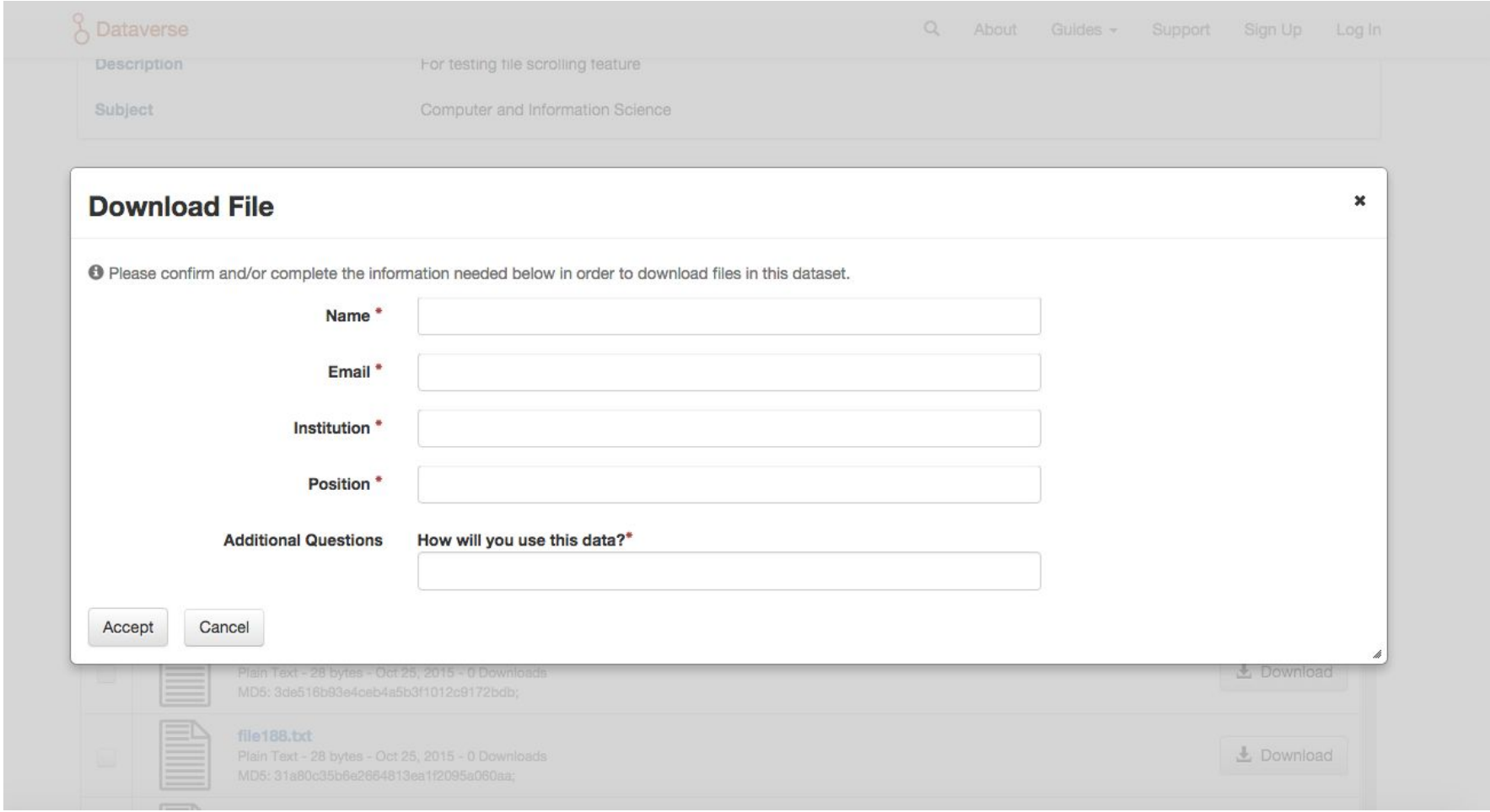
file182.txt
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MD5: 0bf05f11bf1b7f2c20920f4f3e878386; Download

file847.txt
Plain Text - 28 bytes - Oct 25, 2015 - 0 Downloads
MD5: 03c307d5a624101cf2c021fdc5f96a61; Download

file783.txt
Plain Text - 28 bytes - Oct 25, 2015 - 0 Downloads
MD5: 3de516b93e4ceb4a5b3f1012c9172bdb; Download

file188.txt
Plain Text - 28 bytes - Oct 25, 2015 - 0 Downloads
MD5: 31a80c35b6e2664813ea1f2095a060aa; Download

file327.txt


Dataverse
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For testing file scrolling feature
Subject
Computer and Information Science
Download File
Please confirm and/or complete the information needed below in order to download files in this dataset.
Name *
Email *
Institution *
Position *
Additional Questions
How will you use this data?*
Accept
Cancel
Plain Text - 28 bytes - Oct 25, 2015 - 0 Downloads
MD5: 3de516b93e4ceb4a5b3f1012c9172bdb;
Download
file188.txt
Plain Text - 28 bytes - Oct 25, 2015 - 0 Downloads
MD5: 31a80c35b6e2664813ea1f2095a060aa;
Download

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You are like your users

When designing a website, it's ***easy to assume that everybody is like you.*** However, this ***leads to a strong bias and often ends in an inefficient design.***

You evidently know a lot about your services and your website; you're passionate about them. Your users, on the other hand, are likely to not care that much. They have ***different attitudes and goals, and just want to get things done on your website.***

To avoid this bias, ***you need to learn about your users, involve them in the design process, and interact with them.***


WE INTERVIEWED HUNDREDS OF USERS AND TURNED ALL OF THEIR SUGGESTIONS INTO FEATURES.
AS IT TURNS OUT, EVERY USER WE TALKED TO WAS AN IDIOT, AND THEIR DUMB SUGGESTIONS RUINED OUR PRODUCT.
IN HINDSIGHT, WE PROBABLY SHOULD HAVE TALKED TO PEOPLE WHO WORK OUTSIDE THIS BUILDING.
Dilbert.com DilbertCartoonist@gmail.com
5-7-12 ©2012 Scott Adams, Inc./Dist. by Universal Uclick

If the user can’t use it, it doesn’t work.
— Susan Dray

Thank you! Any questions?

Contact: gdurand@iq.harvard.edu or equigley@iq.harvard.edu